MAR-PE-13: Standard Operating Procedure for Cisco PE Node Optical Transport Failure (ALARM_2631)

Node: MAR-PE-13 | **Vendor:** Cisco | **Severity:** Major | **Alarm Code:** ALARM_2631

Description

The MAR-PE-13 node, situated at the Marathahalli Core site, is a critical component of the Cisco optical transport network. As a PE (Provider Edge) node, it plays a pivotal role in ensuring seamless data transmission between the service provider's network and the external clients. The occurrence of an Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) in this node, signified by the ALARM_2631 code, is a major concern that necessitates prompt attention. This failure domain arises due to the inherent limitations of the SFP/XFP transceivers, which are prone to degradation over time, and the physical susceptibility of the fiber infrastructure to cuts and power losses. Furthermore, the complexity of the Cisco optical transport network, which relies on the Optical Transport Network (OTN) protocol, can sometimes lead to issues with signal transmission and reception. The OTN protocol, used for transporting optical signals, is sensitive to changes in the network topology and can be affected by the degradation of SFP/XFP transceivers or fiber cuts.

Pre-requisite Checks

- Ensure that the remote hands are dispatched to the Marathahalli Core site for assistance.

- Verify the status of the redundant link to ensure that it is operational and can take over in case of a failure.

- Check the SFP/XFP transceivers for any signs of degradation or damage.

Potential Root Causes

Cause	Probability
SFP/XFP Transceiver Degradation	High
Fiber Cut	Medium
Rx/Tx Power Loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show controllers t1 0/0/0, Expected Output: T1 0/0/0 is up, line protocol is up, Description: This command will verify the status of the T1 interface on the MAR-PE-13 node. It will check if the line protocol is up, indicating that the interface is functional.

Command: show interface POS 0/0/0, Expected Output: POS 0/0/0 is up, line protocol is up, Description: This command will verify the status of the POS interface on the MAR-PE-13 node. It will check if the line protocol is up, indicating that the interface is functional.
```

```
Command: show log | include ALARM_2631, Expected Output: ALARM_2631: Optical Transport Failure, Description: This command will search the log files for any occurrences of the ALARM_2631 code. It will provide information about the failure, including the time and date it occurred.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP Transceiver, Command: no controllers t1 0/0/0, Description: This command will remove the T1 interface from the MAR-PE-13 node. Once removed, the SFP/XFP transceiver can be replaced with a new one.

Step 2: Step: Verify the Fiber Connection, Command: show interface POS 0/0/0, Description: This command will verify the status of the POS interface on the MAR-PE-13 node. It will check if the line protocol is up, indicating that the interface is functional.

Step 3: Step: Check the Rx/Tx Power Levels, Command: show controllers t1 0/0/0, Description: This command will verify the Rx/Tx power levels on the T1 interface. It will check if the power levels are within the acceptable range.

Post-Check Verification

```
Command: show log | include ALARM_2631, Expected Output: No occurrences of ALARM_2631, Description: This command will search the log files for any occurrences of the ALARM_2631 code. It will verify that the failure has been resolved.

Command: show interface POS 0/0/0, Expected Output: POS 0/0/0 is up, line protocol is up, Description: This command will verify the status of the POS interface on the MAR-PE-13 node. It will check if the line protocol is up, indicating that the interface is functional.
```

Roll-back Procedures

- Step: Revert the SFP/XFP Transceiver Replacement, Command: controllers t1 0/0/0, Description: This command will re-activate the T1 interface on the MAR-PE-13 node. It will revert the changes made during the resolution process.

- Step: Verify the Original Configuration, Command: show running-config, Description: This command will verify the original configuration of the MAR-PE-13 node. It will check if the changes made during the resolution process have been reverted.